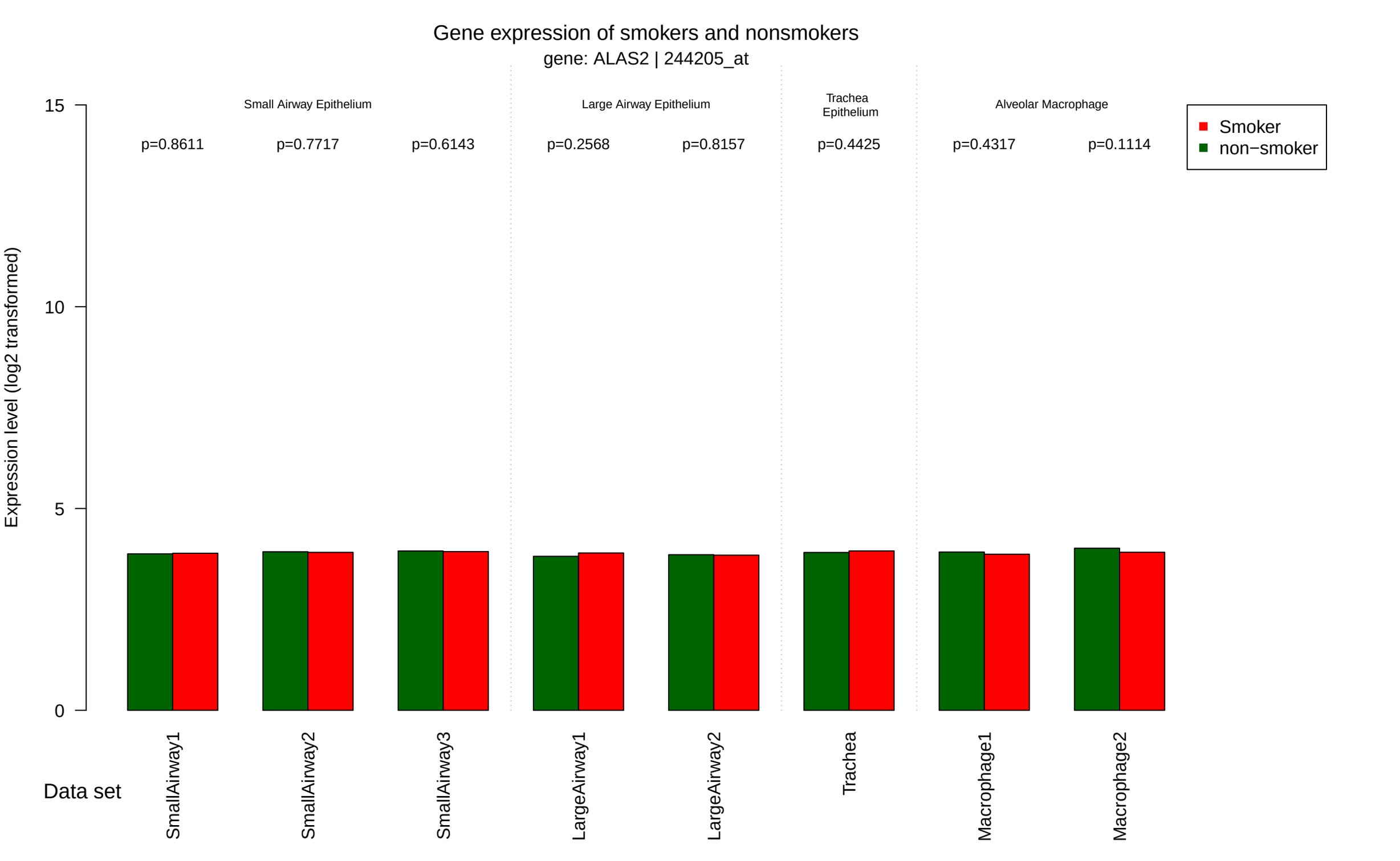
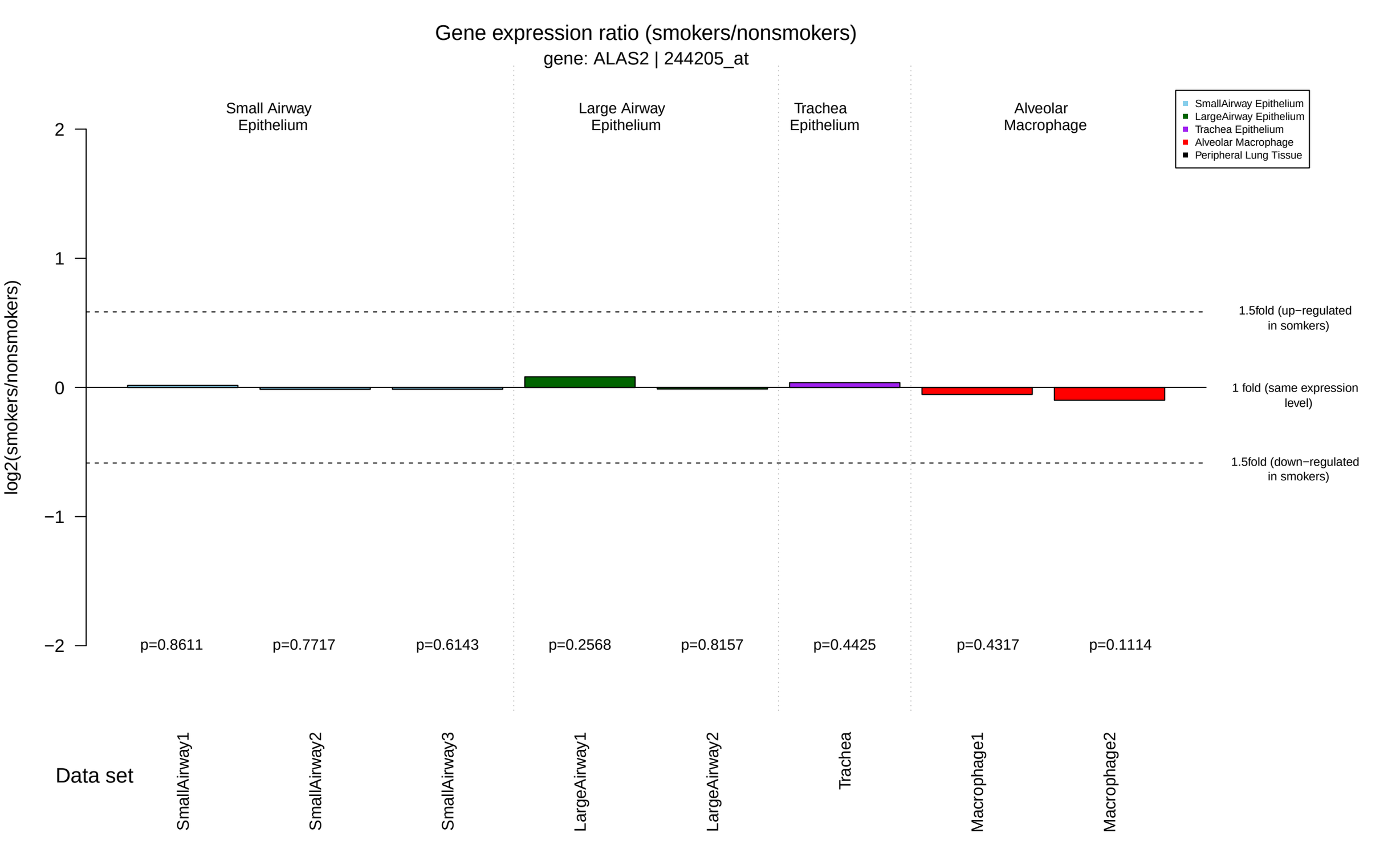
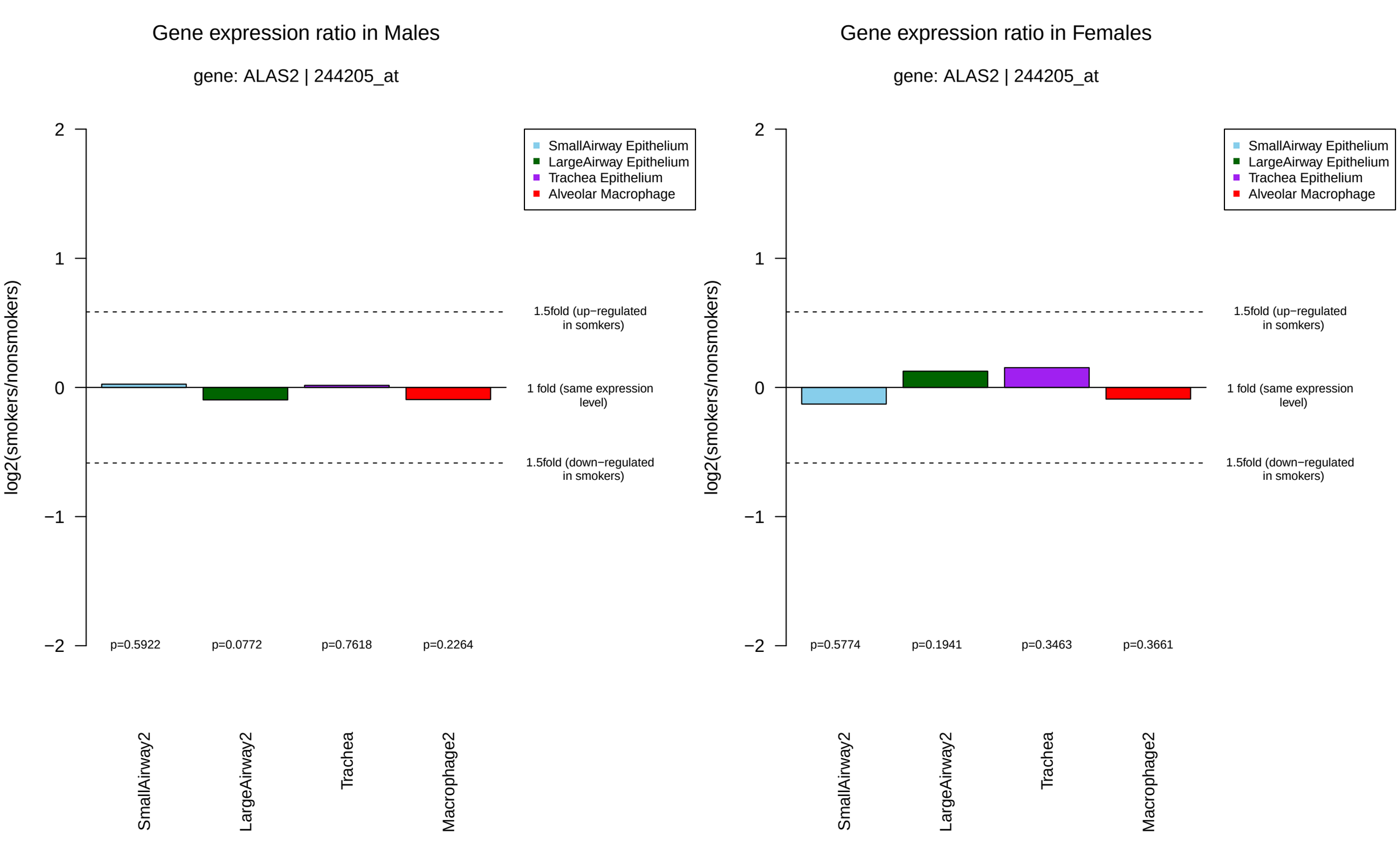
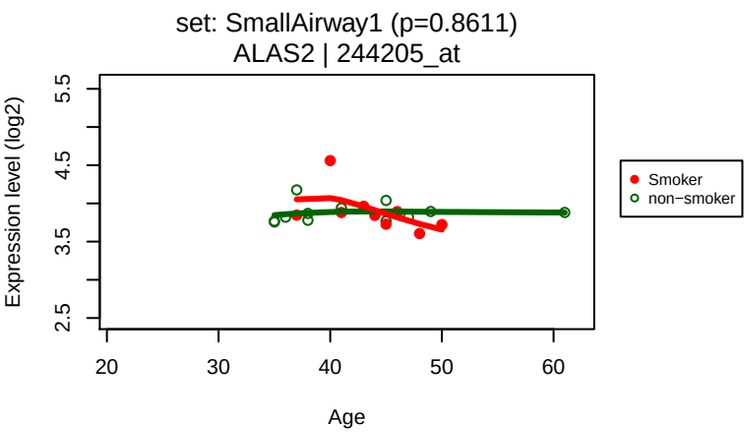
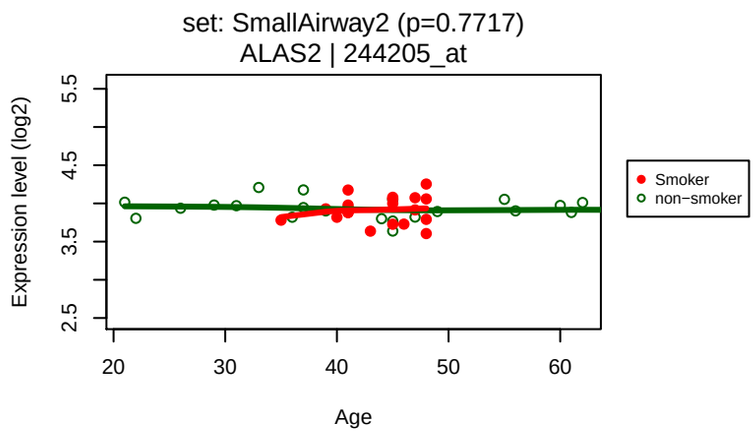
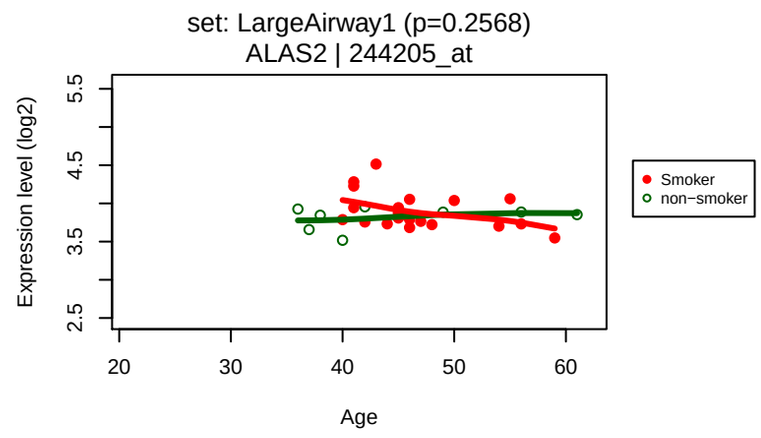

Gene expression of smokers and nonsmokers
gene: ALAS2 | 244205_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8611
p=0.7717
p=0.6143
p=0.2568
p=0.8157
p=0.4425
p=0.4317
p=0.1114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

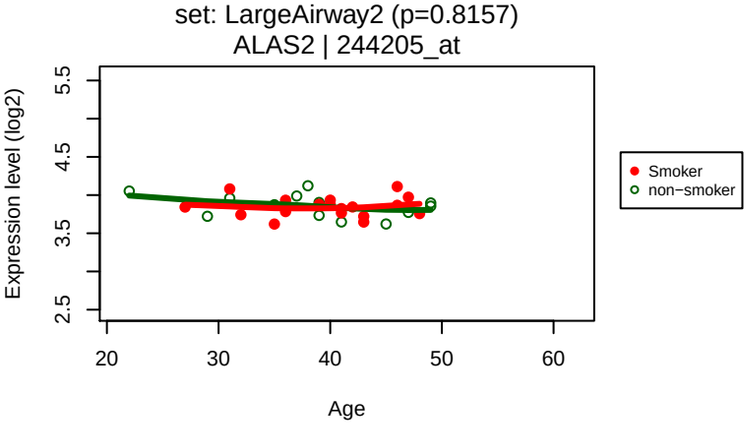

Gene expression ratio (smokers/nonsmokers)
gene: ALAS2 | 244205_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8611
p=0.7717
p=0.6143
p=0.2568
p=0.8157
p=0.4425
p=0.4317
p=0.1114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

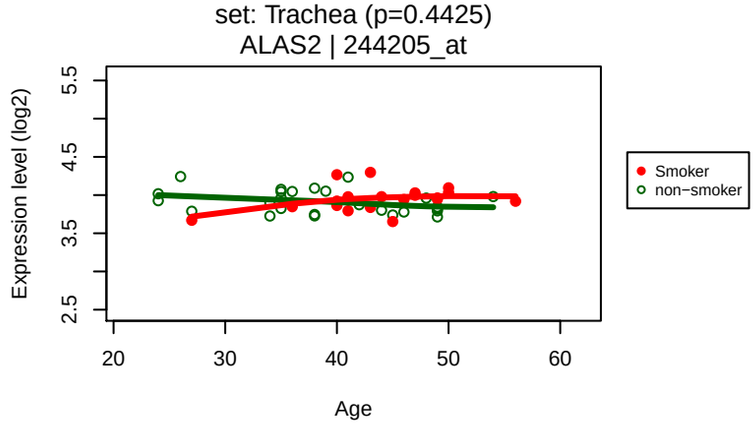

Gene expression ratio in Males
gene: ALAS2 | 244205_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5922
p=0.0772
p=0.7618
p=0.2264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALAS2 | 244205_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5774
p=0.1941
p=0.3463
p=0.3661
SmallAirway2
LargeAirway2
Trachea
Macrophage2

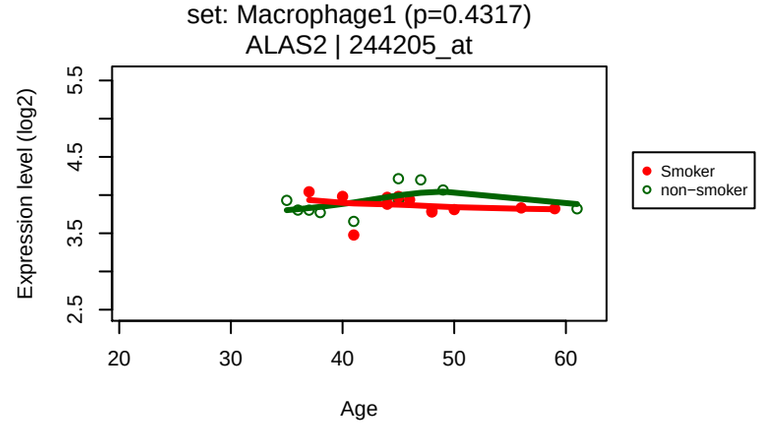

set: SmallAirway1 (p=0.8611)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker

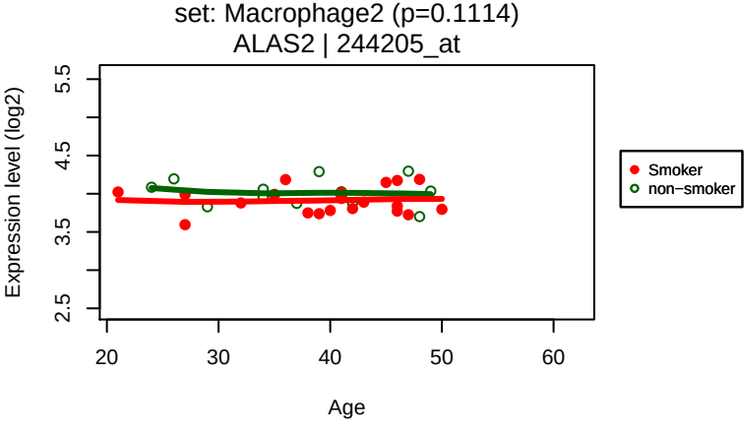

set: SmallAirway2 (p=0.7717)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker

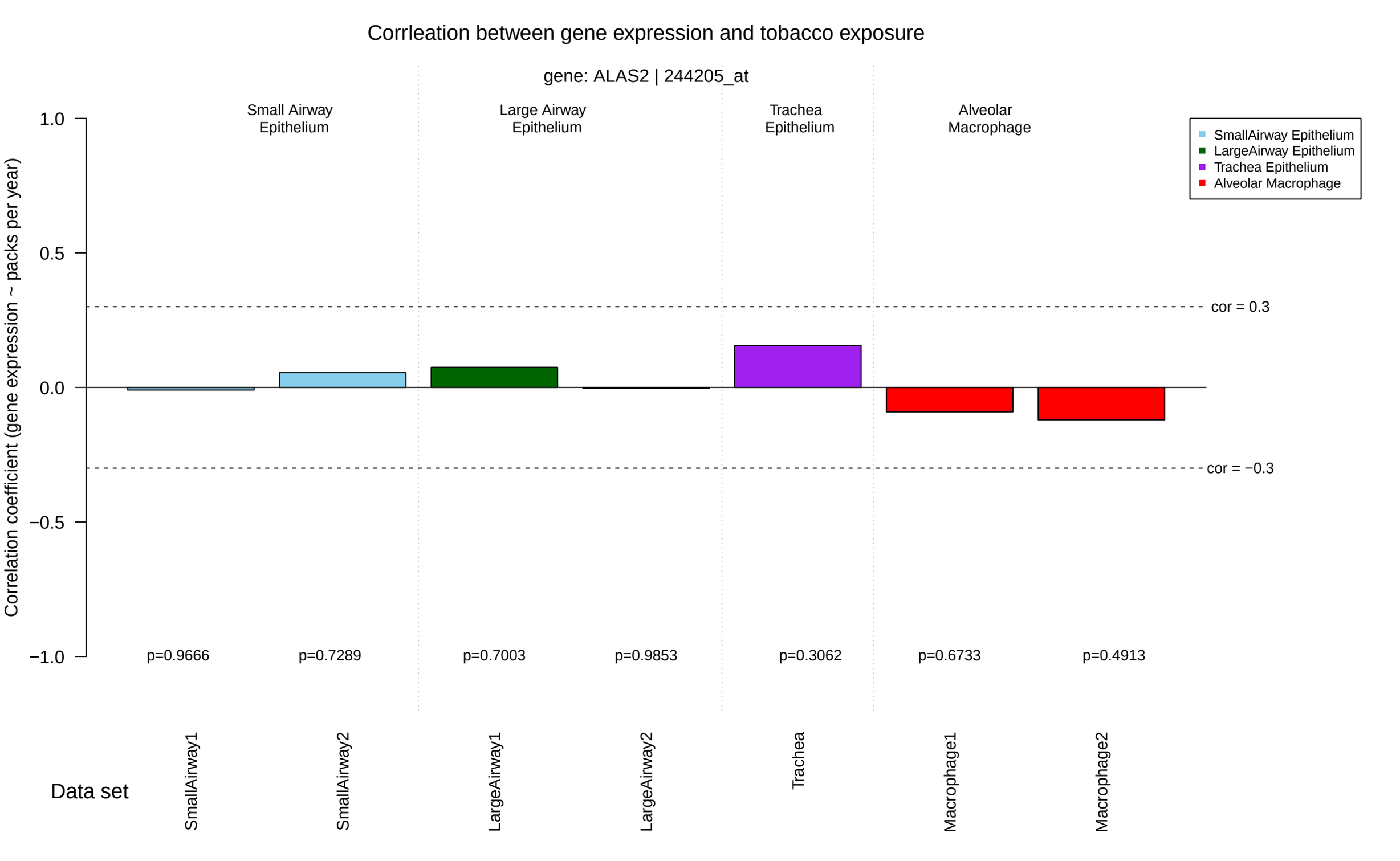

set: LargeAirway1 (p=0.2568)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8157)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4425)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4317)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1114)
ALAS2 | 244205_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALAS2 | 244205_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9666
p=0.7289
p=0.7003
p=0.9853
p=0.3062
p=0.6733
p=0.4913
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2